	PRODUCTION	SEMANTIC RULES
1)	$L \rightarrow E\ \mathbf{n}$	$L.val = E.val$
2)	$E \rightarrow E_1\ +\ T$	$E.val = E_1.val + T.val$
3)	$E \rightarrow T$	$E.val = T.val$
4)	$T \rightarrow T_1\ *\ F$	$T.val = T_1.val \times F.val$
5)	$T \rightarrow F$	$T.val = F.val$
6)	$F \rightarrow (\ E\ )$	$F.val = E.val$
7)	$F \rightarrow \mathbf{digit}$	$F.val = \mathbf{digit}.\text{lexval}$

Figure 5.1: Syntax-directed definition of a simple desk calculator

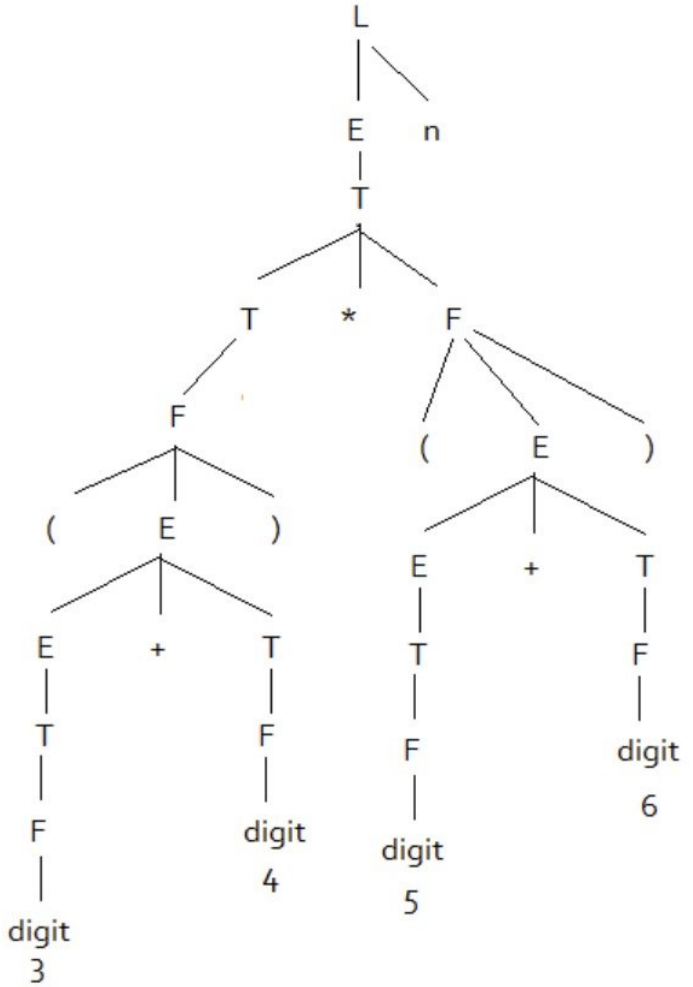
For the SDD(SYNTAX-DIRECTED DEFINITIONS) of Fig. 5.1, give annotated parse trees for the following expressions:

a. $(3 + 4) * (5 + 6)n.$
b. $1 * 2 * 3 * (4 + 5)n.$
c. $(9 + 8 * (7 + 6) + 5) * 4n.$

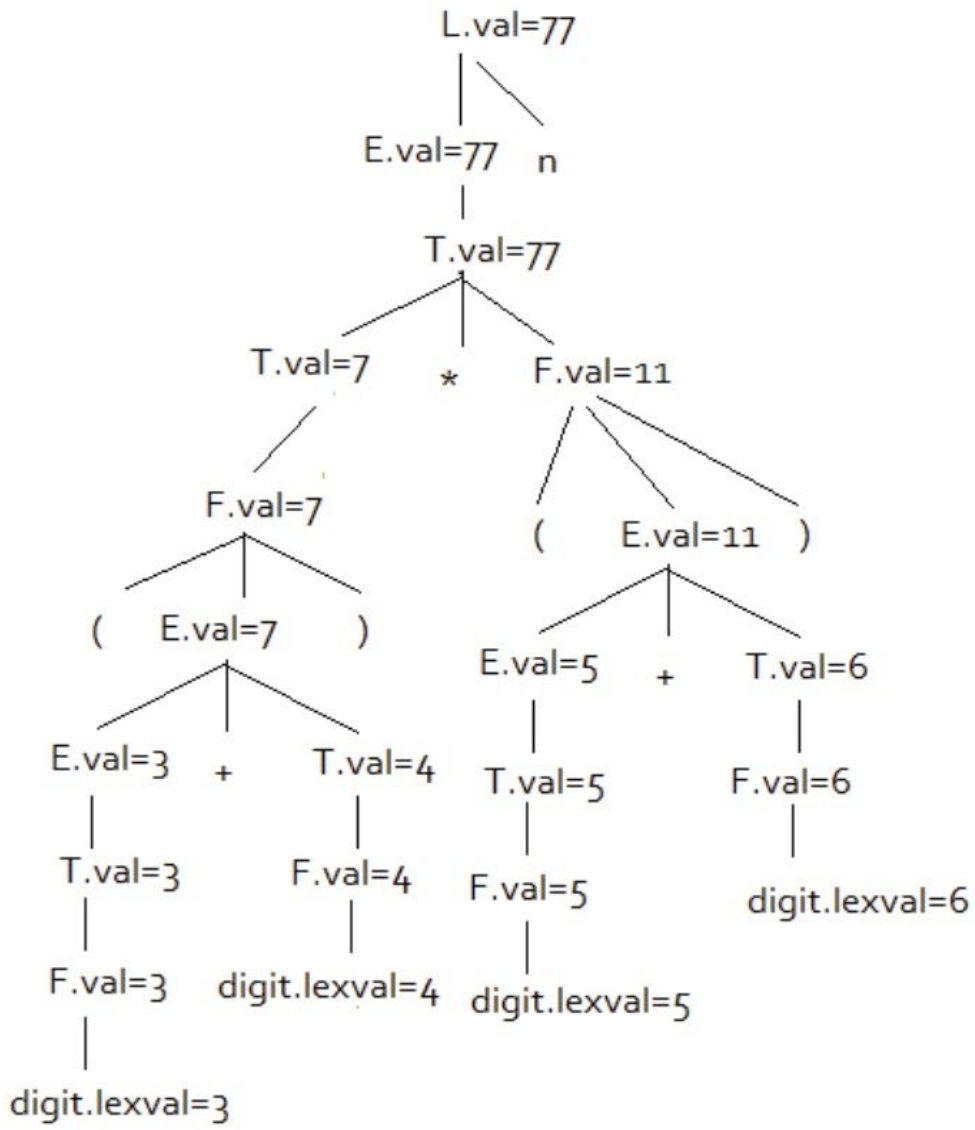
Here all the attributes are synthesized attributes. The attribute value of digit is provided by lexical analyzer since it is a terminal

1. (3+4)∗(5+6)n

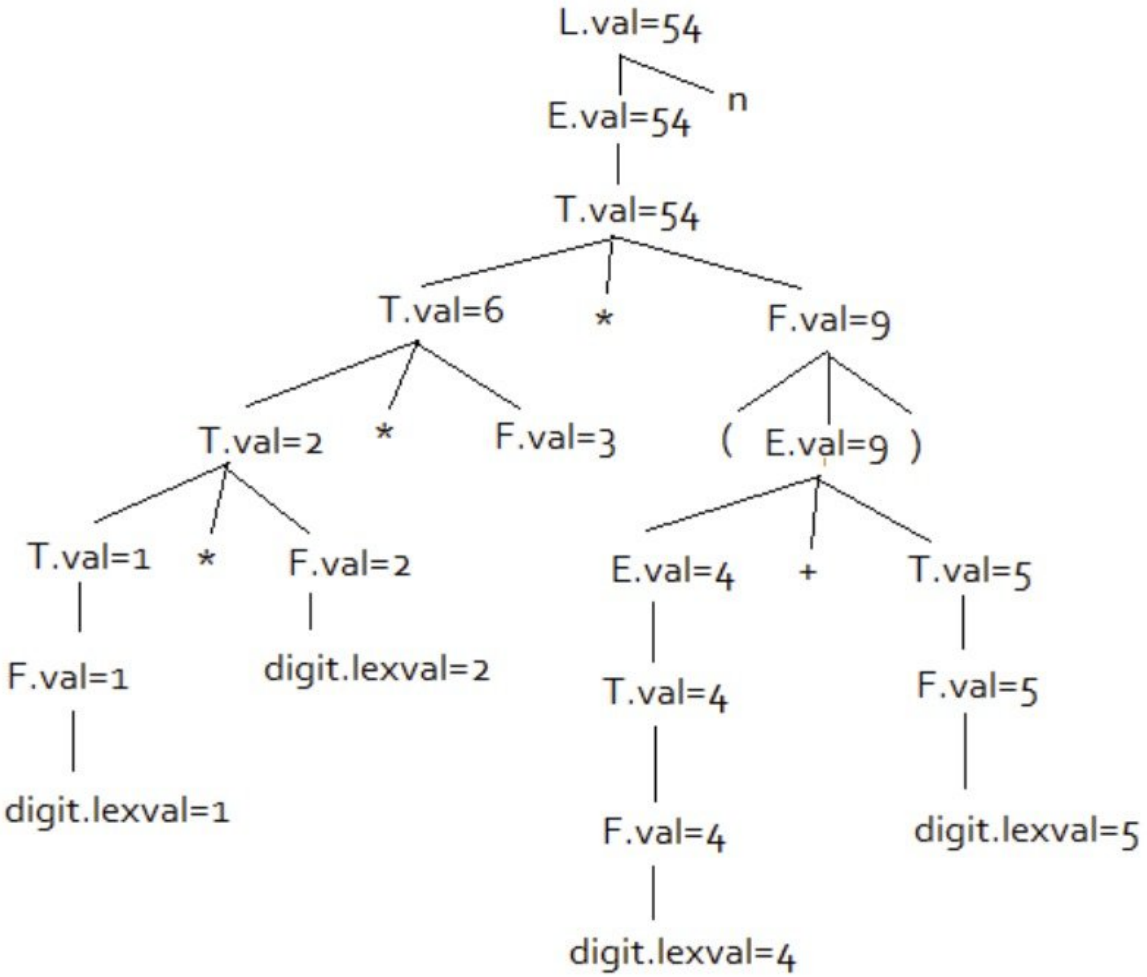
Parse Tree



Annotated Parse Tree

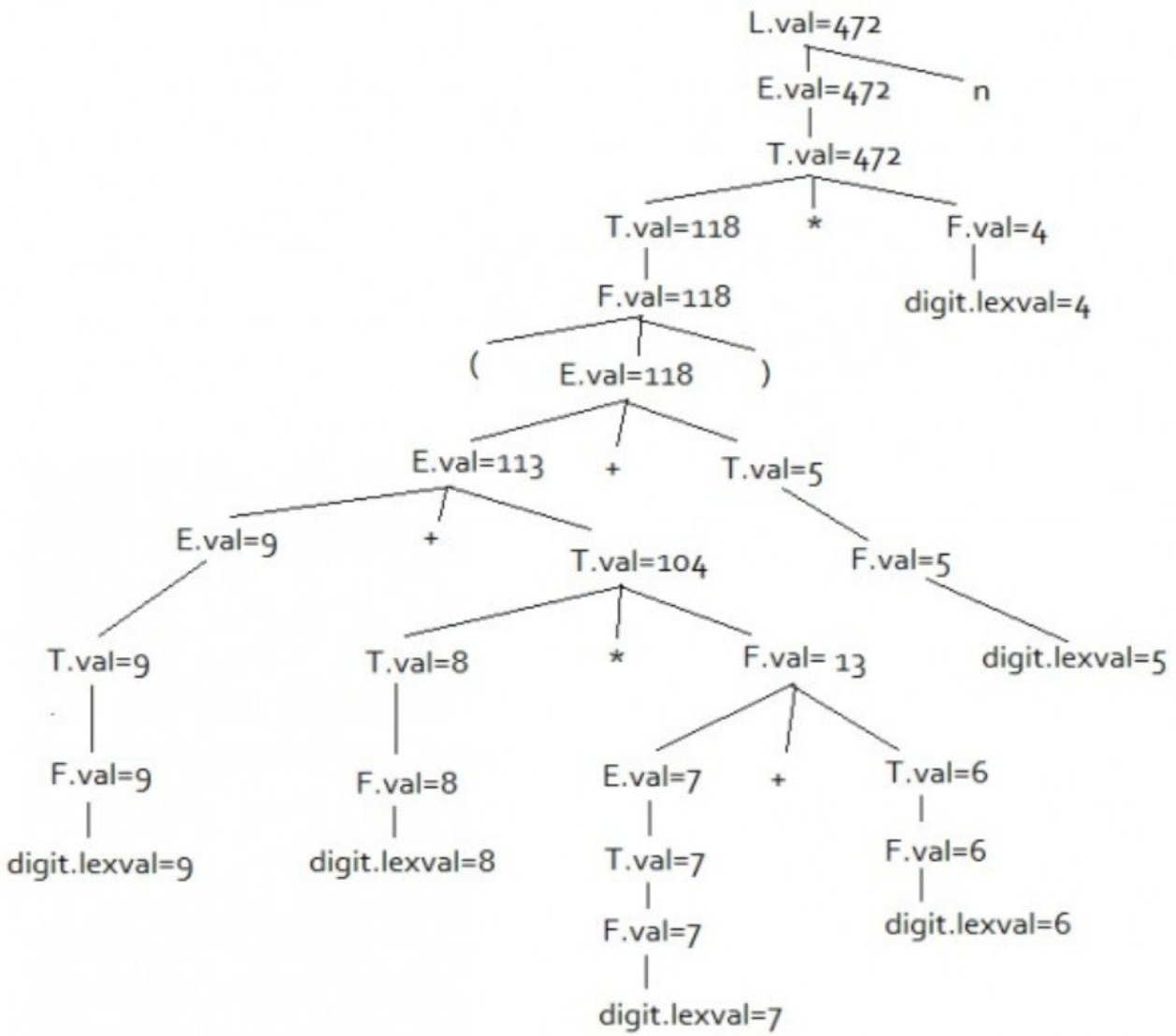


2. 1∗2∗3∗(4+5)n



Synthesized attributes are used in conjunction with bottom up parsers to pass info up the tree

3. (9+8∗(7+6)+5)∗4n


L.val=472
E.val=472
n
T.val=472
T.val=118
*
F.val=4
F.val=118
digit.lexval=4
(
E.val=118
)
E.val=113
+
T.val=5
E.val=9
+
T.val=104
F.val=5
T.val=9
T.val=8
*
F.val= 13
digit.lexval=5
F.val=9
F.val=8
E.val=7
+
T.val=6
digit.lexval=9
digit.lexval=8
T.val=7
F.val=6
F.val=7
digit.lexval=6
digit.lexval=7